

Supervised Research (Law, Economics, and Data Science) 851-0763-00L HS2025

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Topic Selection (5%), due Nov 1st.

Group member: Anna Kravchenko (solo)

Project Topic: **Anomaly Detection for CNP Transactions**

Motivation

Card-not-present (CNP) transactions have scaled with e-commerce and, despite representing a small share of overall volume, they generate material losses and customer friction when misclassified. In production, fraud monitoring operates under two binding constraints: (i) limited reviewer capacity, which forces models to deliver a small number of high-value alerts each day, and (ii) governance requirements for time-aware validation, documentation, and human-interpretable rationales. Classic leaderboard objectives (e.g., ROC-AUC) are weak proxies for this reality: they ignore alert budgets, conflate prevalence shifts, and can look strong even when the top of the ranked list, what teams actually review, underperforms.

This project therefore will treat fraud detection as anomaly ranking with top-K alerting. The focus is on (a) alert quality at realistic review budgets (Precision@K / Recall@K), (b) workload impact (alerts per 1 000 transactions; implied analyst-hours), (c) temporal robustness via strictly chronological splits and drift diagnostics, and (d) reviewer-oriented explanations that make individual alerts defensible.

Methodologically, I will emphasize unsupervised detectors (Isolation Forest, tabular Autoencoder) and include a clearly labeled, calibrated supervised baseline only as a reference, so results remain relevant when labels are sparse or delayed.

Dataset

I will use the IEEE-CIS / Vesta Fraud Detection dataset - real, non-synthetic CNP transactions provided by Vesta Corporation for an official IEEE-CIS challenge. The labeled training data come as two tables: `train_transaction.csv` (transaction features plus the binary target `isFraud`) and `train_identity.csv` (device, browser, and network identity features). They join on **TransactionID**; identity information is present only for a subset of transactions. Time is represented by **TransactionDT**, a delta from a hidden reference timestamp that preserves event order. The schema contains several hundred anonymized variables across well-known families - `ProductCD`, `card1-card6`,

addr1/addr2, P_emaildomain/R_emaildomain, match flags M1-M9, device fields (DeviceType, DeviceInfo), and identity indicators id_12-id_38 - along with numerous engineered numeric features. The training set comprises ~590k transactions and exhibits severe class imbalance, which is characteristic of CNP fraud. The competition test files are unlabeled and will not be used for evaluation; instead, I will build chronological train/validation/test windows inside the labeled training data, ensuring time-respecting model selection and drift assessment.

Data availability: [IEEE-CIS / Vesta Fraud Detection dataset \(official competition release, hosted by IEEE-CIS on Kaggle\)](#). Data provider: [Vesta Corporation](#).

Approach

I will first join `train_transaction` and `train_identity` on `TransactionID`, add explicit missing-value indicators, impute numeric and categorical fields, and standardize continuous features. Categorical variables will be encoded with leakage-safe folds (e.g., frequency/target encoding on the training window only). Transactions will be sorted by `TransactionDT` to reconstruct event order, and I will define strictly chronological train, validation, and test windows. Before modeling, I will profile the data with rich EDA - class imbalance and fraud-rate over time, distributions for key fields, a missingness heatmap - and screen for temporal drift using monthly PSI and stability plots.

Modeling will center on unsupervised anomaly detection: an Isolation Forest and a tabular Autoencoder whose reconstruction error serves as the anomaly score. To contextualize results when labels exist, I will include a single calibrated supervised baseline (LightGBM/XGBoost with Platt or isotonic calibration), clearly marked as a comparator rather than the project's focus. Decision-making follows an alert-budget regime: for each day (derived from `TransactionDT`), transactions are ranked by risk and only the top-K are surfaced for review; I will report results for $K = 50, 100, \text{ and } 200$, and also a fixed alert-rate variant that scales K with daily volume. Evaluation emphasizes operational usefulness- Precision@K and Recall@K for the daily alert list, alerts per 1,000 transactions, and estimated analyst-hours—alongside PR-AUC for overall ranking quality.

I will assess temporal stability and re-calibration needs across time windows and provide explainability via SHAP (global summaries plus short case notes for selected alerts). The final deliverable is a reproducible pipeline with the full figure set (EDA, drift, $\text{Precision@K/Recall@K}$ curves, workload trade-offs, calibration, SHAP) and a concrete threshold/alert-budget recommendation for two staffing scenarios.